Stochastic Gradient Descent

Monday, September 29, 2025 8:37 PM

$$\min f(x)$$

- Assume we only have a noisy version of the gradient

SGD → $$\boxed{x_{k+1} = x_k - \varepsilon_k\left(\nabla f(x_k) + \underset{\uparrow\ \text{noise}}{w_k}\right)}$$

To prove convergence, we make the following assumptions on f:

(i) f is L-smooth, i.e.,

$$f(y) \le f(x) + \nabla^T f(x)(y-x) + \frac{L}{2}\|x-y\|^2$$

(ii) f satisfies the Polyak-Lojasiewicz Condition

$$\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\left(f(x) - f^*\right),$$

where $f^* = \min\limits_x f(x)$

We also make the following assumptions on the noise:

$$E\left(\|w_k\|^2 \mid x_k\right) \le \sigma^2 + \alpha\|\nabla f(x_k)\|^2$$

$$E(w_k \mid x_k) = 0$$

Proof convergence:

$$f(x_{k+1}) \underset{\substack{\uparrow\\ \text{L-smoothness}}}{\le} f(x_k) + \nabla^T f(x_k)(x_{k+1} - x_k) + \frac{1}{2}L\|x_{k+1} - x_k\|^2$$

$$= f(x_k) + \nabla^T f(x_k)\left(-\varepsilon_k \nabla f(x_k) - \varepsilon_k w_k\right) + \frac{1}{2}L\varepsilon_k^2\|f(x_k) + w_k\|^2$$

⇒

$$E\left(f(x_{k+1}) \mid x_k\right) \underset{\substack{\uparrow\\ \text{using assumptions}\\ \text{on the noise}}}{\le} f(x_k) - \varepsilon_k\|\nabla f(x_k)\|^2 + \frac{1}{2}L\varepsilon_k^2\left(\|f(x_k)\|^2 + \sigma^2 + \alpha\|\nabla f(x_k)\|^2\right)$$

$$= f(x_k) - \varepsilon_k\left(1 - \frac{1}{2}L\varepsilon_k(1+\alpha)\right)\|\nabla f(x_k)\|^2 + \frac{1}{2}L\varepsilon_k^2\sigma^2$$

$$\le f(x_k) - 2\varepsilon_k\mu\left(1 - \frac{1}{2}L\varepsilon_k(1+\alpha)\right)\left(f(x_k) - f^*\right) + \frac{1}{2}L\varepsilon_k^2\sigma^2$$

$$\Rightarrow \boxed{E\left(f(x_{k+1}) - f^*\right) \le \left(1 - 2\varepsilon_k\mu\left(1 - \frac{1}{2}L\varepsilon_k(1+\alpha)\right)\right)\left(f(x_k) - f^*\right) + \frac{1}{2}L\varepsilon_k^2\sigma^2}$$

As before, one can set $\varepsilon_k = \varepsilon\ \forall k$ to get an $O(\varepsilon)$ convergence and use time-varying step-size to get $E(f(x_k) - f^*) \to 0$ at a certain rate as a function of k.

Example: $$f(x) = \frac{1}{2}x^TQx + b^Tx + c, \quad Q > 0.$$

$$\nabla f(x) = 0 \Rightarrow Qx^* + b = 0 \text{ or } x^* = -Q^{-1}b$$

- f satisfies the L-smoothness condition obviously

$$f(y) \le f(x) + \nabla^T f(x)(y-x) + \frac{1}{2}(y-x)^TQ(y-x)$$

$$\le f(x) + \nabla^T f(x)(y-x) + \frac{1}{2}\lambda_{\min}(Q)\|y-x\|^2$$

P-L condition

- $$\|\nabla f(x)\|^2 = \|Qx + b\|^2$$

$$= \|Q(x + Q^{-1}b)\|^2$$

$$= (x + Q^{-1}b)^TQQ(x + Q^{-1}b)$$

$$\ge \lambda_{\min}(Q)(x + Q^{-1}b)Q(x + Q^{-1}b)$$

To see why, note that $Q^2 = \sum\limits_i \underset{\substack{\downarrow\\ \text{e'values of } Q}}{\lambda_i^2} v_i \underset{\substack{\searrow\\ \text{e'vectors of } Q}}{v_i^T}$

$$\Rightarrow x^TQ^2x = \sum_i \lambda_i^2(x^Tv_i)(v_i^Tx) \quad \forall x$$

$$\ge \lambda_{\min}\sum_i \lambda_i(x^Tv_i)(v_i^Tx)$$

$$= \lambda_{\min}(x^TQx)$$

- $$f(x) = \frac{1}{2}x^TQx + b^Tx + c$$

$$= \frac{1}{2}(x + Q^{-1}b)^TQ(x + Q^{-1}b) + \frac{1}{2}\left(\cancel{-2b^T}x + b^TQ^{-1}b\right) + \cancel{b^T}x + c$$

$$= \frac{1}{2}(x + Q^{-1}b)^TQ(x + Q^{-1}b) + b^TQ^{-1}b + c$$

and $$f^* = b^TQ^{-1}b + c$$

$$f(x) - f^* = \frac{1}{2}(x + Q^{-1}b)^TQ(x + Q^{-1}b)$$

$$\Rightarrow \boxed{\frac{1}{2}\|\nabla f(x)\|^2 \ge \lambda_{\min}(Q)\left(f(x) - f^*\right)}$$

⇒ P-L condition holds

In fact, one can show that the P-L condition holds even if Q is rank-deficient (i.e., $\lambda_{\min} = 0$) (Homework problem?)

Example: $$f(x) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}x^TQ_ix + b_i^Tx + c_i$$

Choose $i \in \{1, 2, \ldots, N\}$ uniformly at random and let

$$g(x) = (Q_ix + b_i)$$

Let $Q = \frac{1}{N}\sum\limits_{i=1}^{N}Q_i$ and similarly b and c

$$\Rightarrow \nabla f(x) = Qx + b \quad \text{and} \quad Qx^* = -b$$

$$\Rightarrow \nabla f(x) = Q(x - x^*)$$

$$g(x) = \nabla f(x) + \underbrace{Q_ix + b_i - Q(x - x^*)}_{w}$$

$$w = Q_ix^* + b_i - Q(x - x^*) + Q_i(x - x^*)$$

$$= Q_ix^* + b_i + (Q_i - Q)(x - x^*)$$

$$\Rightarrow \|w\|^2 \le 2\|Q_ix^* + b_i\|^2 + 2\|(Q_i - Q)(x - x^*)\|^2$$

$$E(\|w\|^2 \mid x) \le 2E_i\left(\|(Q_ix^* + b_i)\|^2 + 2E(\|Q_i - Q\|_{op}^2\|x - x^*\|^2)\right)$$

$$= \sigma^2 + 2K\|x - x^*\|^2$$

with $\sigma^2 = 2E_i\left(\|Q_ix^* + b_i\|^2\right)$

and $K = E\left(\|Q_i - Q\|_{op}^2\right)$

Recall $$\nabla f(x) = Q(x - x^*)$$

$$\Rightarrow Q^{-1}\nabla f(x) = (x - x^*)$$

$$\Rightarrow \|x - x^*\|^2 = \nabla^T f(x)Q^{-2}\nabla f(x)$$

$$\le \frac{1}{\lambda_{\min}^2(Q)}\|\nabla f(x)\|^2$$

$$\Rightarrow E\left(\|w\|^2\right) \le \sigma^2 + \frac{2K}{\lambda_{\min}^2(Q)}\|\nabla f(x)\|^2$$

□

Comments:

(i) If $\|\nabla f(x) - \nabla f(y)\| \le L\|y - x\|$, then the L-smoothness condition is satisfied. To see this, define

$$g(t) = f(x + t(y-x)) \Rightarrow g(0) = f(x) \text{ and } g(1) = f(y)$$

$$g(1) = g(0) + \int_0^1 g'(t)\,dt$$

$$f(y) = f(x) + \int_0^1 \nabla^T f(x + t(y-x))(y-x)\,dt$$

$$= f(x) + \int_0^1 \left(\nabla^T f(x + t(y-x)) - \nabla^T f(x)\right)(y-x)\,dt + \nabla^T f(x)(y-x)$$

$$\underset{\substack{\uparrow\\ \text{Cauchy-Schwartz + condition on } \nabla f}}{\le} f(x) + \nabla^T f(x)(y-x) + \int_0^1 Lt\|y-x\|^2\,dt$$

$$= f(x) + \nabla^T f(x)(y-x) + \frac{1}{2}L\|y-x\|^2$$

(2) f is said to be strongly convex if $\nabla^2 f(x) \ge mI$ for some $m > 0$

$$\Rightarrow f(y) \underset{\substack{\uparrow\\ \text{Taylor's theorem, for some } z = \alpha x + (1-\alpha)y,\ \alpha \in [0,1]}}{=} f(x) + \nabla^T f(x)(y-x) + \frac{1}{2}(y-x)\nabla^2 f(z)(y-x)$$

$$f(y) \ge f(x) + \nabla^T f(x)(y-x) + \frac{m}{2}\|y-x\|^2 \quad \forall x, y$$

Minimize both sides w.r.t. y:

LHS $= f^*$

RHS:

Differentiating w.r.t. y:

$$\nabla f(x) + m(y - x) = 0$$

$$\Rightarrow y = x - \frac{1}{m}\nabla f(x)$$

$$\Rightarrow \text{RHS} = f(x) - \frac{1}{m}\|\nabla f(x)\|^2 + \frac{\cancel{m}}{2}\frac{1}{m^{\cancel{2}}}\|\nabla f(x)\|^2$$

$$= f(x) - \frac{1}{2m}\|\nabla f(x)\|^2$$

$$\Rightarrow \boxed{\frac{1}{2}\|\nabla f(x)\|^2 \ge m\left(f(x) - f^*\right)}$$

We've used this fact:

If $f(y) \ge g(y)\ \forall y$

$$\Rightarrow f(y) \ge g(y) \ge \underbrace{\min_y g(y)}_{\triangleq\, g^*}$$

$$\Rightarrow f(y) \ge g^* \quad \forall y$$

$$\Rightarrow \min_y f(y) \ge g^* = \min_y g(y)$$